ENGINEERING MATHEMATICS

UNIT–IV : INTEGRAL CALCULUS AND APPLICATIONS

(As per DCET C20 Diploma Curriculum — Marks: 4)

DCET Exam Note:

Syllabus Coverage:

- List of standard integrals and Basic rules of integration
- Evaluation of integrals of simple function and their combination
- Methods of integration
- Concept of definite integrals
- Applications of definite integrals

1. BASIC CONCEPTS

Integration Definition

Integration is the reverse process of differentiation.
If $\frac{d}{dx}F(x) = f(x)$, then:

$$\int f(x)\,dx = F(x) + C$$

where C is the constant of integration.

Important:

Notation:
$\int f(x)\,dx$ = Indefinite integral (with C)
$\int_a^b f(x)\,dx$ = Definite integral (numerical value)
C = Constant of integration (MUST include in indefinite integrals)

2. STANDARD INTEGRALS (MUST MEMORIZE)

Formula:

Basic Integrals:

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} \, dx = \ln|x| + C$$

$$\int e^x \, dx = e^x + C$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1)$$

Formula:

Trigonometric Integrals:

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

Formula:

Other Important Integrals:

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x + C$$

$$\int \frac{1}{1+x^2} \, dx = \tan^{-1} x + C$$

$$\int \frac{1}{\sqrt{x^2+a^2}} \, dx = \ln|x + \sqrt{x^2+a^2}| + C$$

$$\int \frac{1}{\sqrt{x^2-a^2}} \, dx = \ln|x + \sqrt{x^2-a^2}| + C$$

3. BASIC RULES OF INTEGRATION

3.1 Constant Multiple Rule

Formula:

$$\int k f(x) \, dx = k \int f(x) \, dx \quad \text{(where } k \text{ is constant)}$$

3.2 Sum/Difference Rule

Formula:

$$\int [f(x) \pm g(x)]\, dx = \int f(x)\, dx \pm \int g(x)\, dx$$

3.3 Power Rule Extension

Formula:

$$\int (ax+b)^n\, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C \qquad (n \neq -1)$$

Illustration 3.1

Integrate: $\int (3x^2 + 2x - 5)\, dx$

Solution:

$$\int (3x^2 + 2x - 5)\, dx = 3\int x^2\, dx + 2\int x\, dx - 5\int dx$$
$$= 3 \cdot \frac{x^3}{3} + 2 \cdot \frac{x^2}{2} - 5x + C = x^3 + x^2 - 5x + C$$

Illustration 3.2

Integrate: $\int (2x+3)^4\, dx$

Solution: Let $u = 2x + 3$, then $du = 2dx \Rightarrow dx = \frac{du}{2}$

$$\int (2x+3)^4\, dx = \int u^4 \cdot \frac{du}{2} = \frac{1}{2}\int u^4\, du = \frac{1}{2} \cdot \frac{u^5}{5} + C$$
$$= \frac{(2x+3)^5}{10} + C$$

4. METHODS OF INTEGRATION

4.1 Integration by Substitution

Method:

When to use: When integrand contains $f(ax+b)$ or composite function

Steps:

1. Identify substitution: $u = g(x)$
2. Find $du = g'(x)dx$
3. Substitute u and du into integral
4. Integrate with respect to u
5. Substitute back $u = g(x)$

Illustration 4.1

Integrate: $\int x\sin(x^2)\,dx$

Solution: Let $u = x^2$, then $du = 2xdx \Rightarrow xdx = \frac{du}{2}$

$$\int x\sin(x^2)\,dx = \int \sin u \cdot \frac{du}{2} = \frac{1}{2}\int \sin u\,du = -\frac{1}{2}\cos u + C$$
$$= -\frac{1}{2}\cos(x^2) + C$$

4.2 Integration by Parts

Method:

Formula: $\int u\,dv = uv - \int v\,du$

ILATE Rule (Order of choosing u):

- I: Inverse trigonometric (e.g., $\sin^{-1} x$)
- L: Logarithmic (e.g., $\ln x$)
- A: Algebraic (e.g., x^n)
- T: Trigonometric (e.g., $\sin x$)
- E: Exponential (e.g., e^x)

Illustration 4.2

Integrate: $\int xe^x\,dx$

Solution: Let $u = x$, $dv = e^x dx$
Then $du = dx$, $v = e^x$

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C$$

4.3 Integration by Partial Fractions

Method:

For rational functions $\dfrac{P(x)}{Q(x)}$:

- If degree(P) degree(Q): Do long division first
- Factor $Q(x)$ completely
- Write partial fractions based on factors
- Integrate each term separately

Illustration 4.3

Integrate: $\int \dfrac{x+1}{x^2-1}\,dx$

Solution: Factor denominator: $x^2 - 1 = (x-1)(x+1)$
Partial fractions: $\dfrac{x+1}{(x-1)(x+1)} = \dfrac{A}{x-1} + \dfrac{B}{x+1}$
$x + 1 = A(x+1) + B(x-1)$
Let $x = 1$: $2 = 2A \Rightarrow A = 1$
Let $x = -1$: $0 = -2B \Rightarrow B = 0$

$$\int \frac{x+1}{x^2-1}\,dx = \int \frac{1}{x-1}\,dx = \ln|x-1| + C$$

5. DEFINITE INTEGRALS

Fundamental Theorem of Calculus

If $F(x)$ is an antiderivative of $f(x)$, then:

$$\int_a^b f(x)\,dx = F(b) - F(a) = [F(x)]_a^b$$

Important:

Properties of Definite Integrals:

- $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
- $\int_a^a f(x)\,dx = 0$
- $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
- $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
- $\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$

5.1 Evaluation Steps

Method:

To evaluate $\int_a^b f(x)\,dx$:

1. Find indefinite integral $\int f(x)\,dx = F(x) + C$
2. Calculate $F(b) - F(a)$
3. Note: Constant C cancels out!

Illustration 5.1

Evaluate: $\int_0^1 (3x^2 + 2x)\,dx$

Solution:

$$\int (3x^2 + 2x)\,dx = x^3 + x^2 + C$$

$$\int_0^1 (3x^2 + 2x)\,dx = [x^3 + x^2]_0^1 = (1^3 + 1^2) - (0^3 + 0^2) = 2$$

Illustration 5.2

Evaluate: $\int_0^{\pi/2} \sin x\,dx$

Solution:

$$\int \sin x \, dx = -\cos x + C$$

$$\int_0^{\pi/2} \sin x \, dx = [-\cos x]_0^{\pi/2} = (-\cos \frac{\pi}{2}) - (-\cos 0) = (0) - (-1) = 1$$

6. APPLICATIONS OF DEFINITE INTEGRALS

6.1 Area Under Curves

Application:

Area between curve and x-axis:
$A = \int_a^b y \, dx = \int_a^b f(x) \, dx$
Area between two curves:
$A = \int_a^b [f(x) - g(x)] \, dx \quad$ (where $f(x) \geq g(x)$)

Illustration 6.1

Find area between $y = x^2$ and x-axis from $x = 0$ to $x = 2$

Solution:

$$A = \int_0^2 x^2 \, dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3} \text{ sq units}$$

Illustration 6.2

Find area between $y = x^2$ and $y = x$ from $x = 0$ to $x = 1$

Solution: Intersection: $x^2 = x \Rightarrow x(x-1) = 0 \Rightarrow x = 0, 1$
In $[0, 1]$, $x \geq x^2$

$$A = \int_0^1 (x - x^2) \, dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \left(\frac{1}{2} - \frac{1}{3}\right) - 0 = \frac{1}{6} \text{ sq units}$$

6.2 Volume of Revolution

Application:

Disk Method (rotation about x-axis):
$V = \pi \int_a^b [f(x)]^2 \, dx$
Washer Method (between two curves):
$V = \pi \int_a^b [f(x)^2 - g(x)^2] \, dx$

Illustration 6.3

Find volume when region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$ is rotated about x-axis.

Solution:

$$V = \pi \int_0^4 (\sqrt{x})^2 \, dx = \pi \int_0^4 x \, dx = \pi \left[\frac{x^2}{2} \right]_0^4$$

$$= \pi \left(\frac{16}{2} - 0 \right) = 8\pi \text{ cubic units}$$

6.3 Mean Value of Function

Formula:

ean value of $f(x)$ over $[a, b]$:

$$\text{Mean value} = \frac{1}{b-a} \int_a^b f(x) \, dx$$

Illustration 6.4

Find mean value of $f(x) = 2x + 1$ over $[0, 2]$

Solution:

$$\int_0^2 (2x+1) \, dx = [x^2 + x]_0^2 = (4+2) - 0 = 6$$

$$\text{Mean value} = \frac{1}{2-0} \times 6 = 3$$

7. SPECIAL INTEGRALS

7.1 Trigonometric Identities

Formula:

seful identities for integration:

- $\sin^2 x = \dfrac{1 - \cos 2x}{2}$
- $\cos^2 x = \dfrac{1 + \cos 2x}{2}$
- $\sin 2x = 2 \sin x \cos x$
- $1 + \tan^2 x = \sec^2 x$

Illustration 7.1

Integrate: $\int \sin^2 x \, dx$

Solution:

$$\int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx = \frac{1}{2} \int dx - \frac{1}{2} \int \cos 2x \, dx$$

$$= \frac{x}{2} - \frac{1}{4} \sin 2x + C$$

7.2 Integration of Even/Odd Functions

Important:

For symmetric limits $[-a, a]$:

- If $f(x)$ is even: $\int_{-a}^{a} f(x) \, dx = 2 \int_{0}^{a} f(x) \, dx$
- If $f(x)$ is odd: $\int_{-a}^{a} f(x) \, dx = 0$

Even function: $f(-x) = f(x)$ (symmetric about y-axis)

Odd function: $f(-x) = -f(x)$ (symmetric about origin)

Practice Problems (DCET Pattern)

DCET Exam Note:

Solve these typical DCET questions:

1. $\int (4x^3 - 3x^2 + 2)\, dx$ [Ans: $x^4 - x^3 + 2x + C$]
2. $\int (2x-1)^3\, dx$ [Ans: $\frac{(2x-1)^4}{8} + C$]
3. $\int x\cos(x^2)\, dx$ [Ans: $\frac{1}{2}\sin(x^2) + C$]
4. $\int \ln x\, dx$ [Ans: $x\ln x - x + C$]
5. $\int \dfrac{dx}{x^2+4}$ [Ans: $\frac{1}{2}\tan^{-1}(\frac{x}{2}) + C$]
6. $\int_0^1 (x^2+1)\, dx$ [Ans: $\frac{4}{3}$]
7. $\int_0^\pi \cos x\, dx$ [Ans: 0]
8. Area between $y = 4 - x^2$ and x-axis [Ans: $\frac{32}{3}$ sq units]
9. Volume when $y = x$ from 0 to 2 rotated about x-axis [Ans: $\frac{8\pi}{3}$]
10. Mean value of $f(x) = x^2$ on $[0, 3]$ [Ans: 3]

Quick Revision Points

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1)$
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int e^x dx = e^x + C$
- $\int f(ax+b)dx = \frac{F(ax+b)}{a} + C$
- Integration by parts: $\int u dv = uv - \int v du$
- ILATE rule for choosing u
- $\int_a^b f(x)dx = F(b) - F(a)$
- Area = $\int_a^b y dx$
- Volume = $\pi \int_a^b y^2 dx$ (disk method)
- Even function: $\int_{-a}^a f = 2\int_0^a f$
- Odd function: $\int_{-a}^a f = 0$
- Mean value = $\frac{1}{b-a}\int_a^b f(x)dx$

DCET Exam Note:

DCET Exam Tips:

- Memorize all standard integrals
- Always add $+C$ for indefinite integrals
- For definite integrals, C cancels out
- Practice substitution method thoroughly
- Use ILATE rule for integration by parts
- For area problems: Sketch graph if possible
- Check if function is even/odd for symmetric limits
- Volume problems: Remember π in formula
- Partial fractions: Factor denominator completely
- Trigonometric integrals: Use identities to simplify
- Time management: 3-4 minutes per integration problem

Integral Calculus Formula Sheet

Standard Integrals:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

Rules:

$$\int kf(x)dx = k\int f(x)dx$$

$$\int [f \pm g]dx = \int f dx \pm \int g dx$$

$$\int f(ax+b)dx = \frac{F(ax+b)}{a} + C$$

Methods:

Substitution: $\int f(g(x))g'(x)dx = \int f(u)du$

By Parts: $\int udv = uv - \int vdu$

ILATE: Inverse, Log, Algebraic, Trig, Exponential

Applications:

$$\text{Area} = \int_a^b f(x)dx$$

$$\text{Volume} = \pi \int_a^b [f(x)]^2 dx$$

$$\text{Mean} = \frac{1}{b-a}\int_a^b f(x)dx$$